Lab 3 Assignment

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Answer:

1. BFS is designed to find the path with the fewest number of hops or edges from the start node to the goal node. It expands nodes level by level and does not consider edge costs or distances. Therefore, even if distance information is available, BFS ignores it because its objective is to minimize the number of flights, not the travel distance.
2. The two main changes are:
   - The frontier changes from a Queue (FIFO) to a Stack (LIFO).
   - The operation used to remove nodes changes:
     BFS: popleft()
     DFS: pop()

   The pop() operation makes DFS order dependent because DFS always explores the most recently added neighbor first. Different neighbor orders can therefore produce different search paths.
3. A goal state might be generated through a more expensive route before a cheaper route is discovered. If UCS stopped immediately when generating the goal, it could return a non-optimal solution. Waiting until the goal is popped from the priority queue guarantees that the cheapest path has been found.
4. Use expanded.
   Reason:
   - expanded measures the actual work performed by the algorithm.
   - runtime_ms depends on hardware, processor speed, memory, and system load.
   - Therefore, expanded provides a fair comparison across different computers.
5. GBFS (Greedy Best-First Search) ignores: $g(n)$
   which is the total cost from the start node.
   While A* uses: $f(n) = g(n) + h(n)$
   Since GBFS only considers the heuristic value $h(n)$, it may choose routes that appear closer to the goal but have a higher overall cost. This explains why GBFS can produce suboptimal paths while A* often finds optimal ones.
6. For UCS,
   Uses only $g(n)$
   X = 8
   Y = 3
   UCS expands Y first.

   For GBFS
   Uses only $h(n)$
   X = 2
   Y = 6
   GBFS expands X first.

   For A*
   Uses $f(n) = g(n) + h(n)$
   X = 8 + 2 = 10

Y = 3 + 6 = 9
Answer: A* expands Y first.

7. A* adds the actual path cost $g(n)$and heuristic cost $h(n)$:
$$f(n) = g(n) + h(n)$$
If they use different units, the value of $f(n)$becomes meaningless. For example, adding kilometers and dollars together would not produce a valid cost estimate. Therefore, both must use the same unit of measurement.
8. Using the reversed graph allows Dijkstra's algorithm to calculate the shortest distance from every node to the goal in a single execution.
Benefits:
   - More efficient than running Dijkstra from every node individually.
   - Produces exact heuristic values for all states.
   - Ensures the heuristic remains admissible and optimal.

9. The SIN→EZE route likely has a very clear best route with little competition from alternative paths. Because the optimal path is already strongly preferred, increasing the heuristic weight to 1.5 does not change the chosen route. Therefore, both standard A* and Weighted A* produce the same result.